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Upper-Level Jet Streaks

This is a sample lesson page from the [Certificate of Achievement in Weather Forecasting](#) offered by the Penn State Department of Meteorology. Any questions about this program can be directed to: [Steve Seman](#)

Prioritize...

When you've finished with this page, you should be able to discuss the role that upper-level jet streaks can play in outbreaks of deep, moist convection. Namely, you should be able to apply the four-quadrant model of a straight jet streak, as well as a "two-quadrant" model of a cyclonically curved jet streak (identifying areas of convergence and divergence aloft, as well as their potential impacts).

Read...

In the spirit of 500-mb shortwave troughs, upper-level jet streaks help to prime the local environment for deep, moist convection. If nothing else, upper-level jet streaks promote high-altitude cooling via upward motion associated with pockets of upper-level divergence. Such cooling aloft increases CAPE by moving the environmental temperature sounding to the left on a skew-T. By the way, "upper-level", in the context of jet streaks, typically means 300 mb during the cold season and 250 mb during the warm season. I'll stick with 300 mb here just to make life simpler.

How do we assess upper-level divergence produced by 300-mb jet streaks? Previously, you learned about the good, old [four-quadrant model](#). Remember? This four-quadrant model holds that upper-level divergence occurs in the left-exit and right-entrance regions of 300-mb jet streaks. But, there's a problem with this model--it's highly idealized, and assumes that jet streaks are "straight" (no curvature in the flow).

In the real world, those assumptions aren't realistic. In fact, [most upper-level jet streaks are curved](#). Even the jet streaks that look pretty straight aren't *perfectly* straight. In contrast to the idealized world of the four-quadrant model, the right-entrance and left-exit regions associated with *real-life* jet streaks are not sure bets for deep, moist convection.

In real life, the right-entrance and left-exit regions are the most statistically favored quadrants for severe weather, but there are no guarantees. In fact, many an outbreak of severe weather has occurred in the right-exit region of an upper-level jet streak. To understand how this quadrant can be favorable for the development of severe weather, we have to go beyond the simple four-quadrant model of straight jet streaks that you're already familiar with, and think about jet streaks from a different perspective that involves vorticity.

Curvature in the flow within a curved jet streak has some implications for vorticity that changes the patterns of convergence and divergence so that they don't nicely match those from our original four-quadrant model. That's the bottom line, but I'll save the details of the vorticity treatment of jet streaks for the *Explore Further* section below, if you're interested. Ultimately, however, computer modeling of cyclonically curved jet streaks suggests that a two-quadrant model is more appropriate than any four-quadrant model, as summarized in the schematic below.

Schematic showing the two-quadrant model of a cyclonically curved jet streak.

The two-quadrant model of a cyclonically curved jet streak (shades of blue indicate wind speeds around its core). Note that upper-level divergence and the associated upward motion in the left-exit region can "bleed" into the right-exit region.

Credit: David Babb

Just like straight jet streaks, divergence and upward motion occur in the left-exit regions of cyclonically curved jet streaks. Similarly, convergence and downward motion are hallmarks of the left-entrance regions. But, there

are also important differences between the two models. Indeed, upper-level divergence and the associated upward motion “bleed” into the adjacent right-exit region of a cyclonically curved jet streak. Similarly, upper-level convergence and the associated sinking motion can also “bleed” into the right-entrance region.

The bottom line is that with a cyclonically curved jet streak, the right-exit region may be favorable for divergence and upward motion, both of which can help prime the atmosphere for thunderstorm development. So, the right-exit region is not off limits to severe weather. Not by a long shot!

Since no single conceptual model works for all jet streaks, it can be challenging for a forecaster to determine exactly when a jet streak becomes curved enough that the right-exit region becomes favorable for upper-level divergence and upward motion, or exactly where the favorable region of upper-level divergence stops. How does a forecaster deal with these uncertainties?

Personally, I think a somewhat unconventional approach is wise. Let's assume that there's an upper-level jet streak in the vicinity of a region where ingredients in the lower troposphere appear to be coming together for an outbreak of severe thunderstorms. Focusing my attention on the corresponding quadrant of the 300-mb jet streak (above the region of favorable ingredients at low levels), I look for reasons why this specific quadrant might *not* be favorable for the development of storms. In other words, I automatically assume from the get-go that this quadrant will support deep, moist convection. Then I look for reasons why it might not be favorable. If I can't find any good reasons why it won't be favorable, then I assume that it will help favor deep, moist convection.

That approach might seem odd to you, but as long-time, renowned severe weather forecaster, Jack Hales, likes to say, "People have died in the wrong jet quadrant." By this he means that low-level uplift and any subsequent eruption of severe thunderstorms are not always constrained to occur in the two most statistically favored quadrants (right-entrance and left-exit regions). To see an example, check out the case from April 26, 1991 in the *Case Study* box below. It's a prime example supporting Jack's sobering observation, as 72 tornadoes killed 24 people on this date.

Case Study...

Severe weather reports from April 26, 1991.

The storm reports on April 26, 1991. Red corresponds to a tornado report, blue to a wind report, and green to a report of large hail (back then, the threshold was 0.75 inches in diameter).

Credit: Storm Prediction Center

April 26, 1991

One aspect of the horrific April 26, 1991 outbreak may have perplexed folks who were only familiar with the four-quadrant model of straight jet streaks. The severe weather in this outbreak occurred in the right-exit region of a cyclonically curved 300-mb jet streak.

First, to give you some basic synoptic background about the case, the [21Z reanalysis of 500-mb heights](#) on April 26, 1991, showcases a strong, negatively tilted trough pivoting eastward over the Central U.S. At the surface, a 992-mb low was centered over Nebraska, with a lee trough extending southward over the western high Plains ([21Z reanalysis of mean sea-level isobars](#)). As you've learned, lee troughs can aid in the formation of dry lines, and that's what happened here. The [21Z reanalysis of two-meter dew points](#) shows a very large gradient in the vicinity of the lee trough. Unfortunately, this dew-point reanalysis uses Kelvins (I don't know why), so I gave you a couple of Fahrenheit markers so that you can more easily pick out the dry and moist air masses. Without reservation, the Gulf of Mexico was open for business as a tongue of relatively high dew points stretched from the Gulf across eastern Kansas and Oklahoma.

Given that Kansas and Oklahoma were hardest hit during the tornado outbreak, here's the [close-up 21Z map of surface station models](#) (4 P.M. local time) that also shows the position of the dry line across these two states. The convergence along this surface boundary provided lift that helped get parcels to the LFC.

Furthermore, the presence of a mid-level jet (note the [wind maximum exceeding 35 meters per second \(roughly 70 knots\) at 500 mb](#)) favored strong vertical wind shear in the layer from the surface to six kilometers, which supports organized, sustained thunderstorms. Additionally, the presence of a [low-level jet stream evident at 850 mb](#) helped to dramatically boost vertical wind shear in the lowest kilometer of the troposphere, which favors rotating updrafts and possibly the development of tornadoes.

So, what we've seen of the big picture on this date seemed ripe for an outbreak of supercells, and possibly tornadoes. What about the 300 mb pattern? Check out the 21Z reanalysis of 300-mb vector winds below.

21Z reanalysis of 300-mb vector winds on April 26, 1991.

The reanalysis of 300-mb vector winds at 21Z on April 26, 1991 (wind speeds color-coded in meters per second). At the time, the core of a curved jet streak, with wind speeds exceeding 47.5 meters per second (92 knots), was centered over New Mexico, placing Kansas, Oklahoma, and Nebraska in the right-exit region.

Credit: Earth System Research Laboratory

Oklahoma, Kansas, and Nebraska were located in the right-exit region of a cyclonically curved 300-mb jet streak (the core of the streak was over New Mexico). It's very likely that some upper-level divergence bled into the right-exit region of the jet streak, further priming the atmosphere for deep, moist convection.

So, the right-exit region of a cyclonically curved jet streak is not off limits to severe weather. As a budding mesoscale forecaster, keep this example in mind whenever you go through your checklist for getting a sense of the overall synoptic-scale pattern. Analyzing 250 mb or 300-mb winds should certainly be part of your routine. Be sure to look for jet streaks and note whether they're primarily straight, or cyclonically curved. Don't rule out thunderstorm development even in right-exit region if the jet streak is cyclonically curved!

If you would like to learn a bit more about this outbreak, you may be interested in the following links:

- [Event Summary](#) from the Oklahoma Climatological Survey
- [YouTube video about the outbreak](#), compiled from segments which originally aired on KWCH-TV in Wichita, Kansas.

Another interesting aspect of this outbreak (which perhaps you did not notice) is that that an upper-level jet streak and a low-level jet stream both played a role. As it turns out, their impact on the same region was not a coincidence. Indeed, the low-level jet stream and the upper-level jet streak (embedded in the mid-latitude jet stream) were *coupled*. We'll explore this new concept in the next section.

Explore Further...

To understand why curved jet streaks behave differently than straight ones, we need to start by thinking about straight jet streaks in a slightly different way. As it turns out, there's a classic pattern of 300-mb absolute vorticity associated with straight jet streaks. While absolute vorticity is the sum of curvature vorticity, shear vorticity, and earth vorticity, we're going to ignore earth vorticity here (it only depends on latitude). We're only interested in curvature vorticity and shear vorticity, and if we're dealing with a straight jet streak, we can eliminate curvature vorticity (because the jet streak is straight--it has no curvature).

Schematic depicting horizontal wind shear in a straight jet streak.

An idealized schematic of a 300-mb jet streak. The various shades of blue represent the 300-mb wind speeds (in knots) around the core of the 300-mb jet streak. The resulting horizontal wind shear causes two "fans" placed just to the north and south of the core of the jet streak to turn counterclockwise and clockwise, respectively.

Credit: David Babb

With earth vorticity and curvature vorticity off the table, establishing the pattern of absolute vorticity in the vicinity of a straight jet streak boils down to shear vorticity, as shown in the schematic of the straight, west-to-east 300-mb jet streak on the right. We assume that the jet streak resides in the Northern Hemisphere.

The various shades of blue represent the 300-mb wind speeds in the core of the 300-mb jet streak. The three wind vectors on the left qualitatively depict the horizontal wind shear associated with the jet streak (the length of each vector indicates the corresponding 300-mb wind speed). When two "fans" are placed just to the north and south of the core of the jet streak, the horizontal wind shear essentially causes the fan north of the jet streak's axis to turn counterclockwise (cyclonically). Similarly, the fan south of the jet streak's axis turns clockwise (anticyclonically). If we add earth vorticity back into the mix, we discover that there is a [vorticity maximum \(vort max\) north of the jet streak's core, and a vorticity minimum \(vort min\) to its south](#).

If an air parcel at the center of the vort max moves eastward toward the left-exit region, it crosses isovorts with lower values. Given that the parcel tries to stay in equilibrium with its environment, it loses some of its cyclonic spin. In short, the parcel's area increases in response to its environment. In other words, there is mass divergence.

The idealized four-quadrant model of a 300-mb jet streak from a vorticity perspective.

The idealized four quadrant model of a 300-mb jet streak from a vorticity perspective. Parcels flowing north and south of the core flow through a vort max (north) and vort min (south), which yields the familiar patterns of upper-level convergence and divergence.

Credit: David Babb

If an air parcel at the center of the vort min moves eastward toward the right-exit region, it crosses isovorts with higher values. In short, the parcel's area decreases in response to its environment, so there is mass convergence. We can make similar arguments for the left- and right-entrance regions. Any way you slice it, you arrive at the now familiar four-quadrant model for a straight jet streak (see schematic on the left).

But, all we've done so far is rebuild the basic four-quadrant model of a straight jet streak using absolute vorticity. The resulting patterns of convergence and divergence are the same as those you've learned before. When a jet streak becomes curved, the added curvature vorticity changes things a bit.

In this example, we followed an air parcel along and observed the changes it underwent (in terms of cyclonic spin and surface area) while moving away from the vort max into the left-exit region. This approach of following an air parcel as it moves along is formally called a "Lagrangian" approach. But, there's another way to look at the same situation without hitching a ride with an air parcel. We can sit tight at a given location downwind of a vort max and watch air parcels as they go by, which is formally referred to as a "Eulerian" approach.

A "Eulerian" approach (where we're fixed in space, watching air parcels pass by us) allows us to observe the approaching advection of absolute vorticity by the 300-mb wind. Eventually, the wind advects higher values of absolute vorticity over our location (as the vort max approaches). So, in time, the 300-mb absolute vorticity increases over our location. In light of this increase in absolute vorticity with time, meteorologists describe this process as **positive vorticity advection** (PVA, for short).

In general, [PVA typically occurs just east of the 300-mb vort max](#) (at point P in the schematic). In most cases, PVA and upper-level divergence go hand in hand, as they do in the left-exit region of a straight 300-mb jet streak. Similarly, **negative vorticity advection** (NVA) at 300 mb often corresponds to upper-level convergence. So there's [NVA and upper-level convergence in the right-exit region of a straight 300-mb jet streak](#) (at Point Q on this schematic).

However, when a jet streak is cyclonically curved, these patterns of positive and negative vorticity advection get distorted somewhat, because of the curvature of the flow. Some positive vorticity advection likely occurs in the right-exit region of a cyclonically curved jet streak, which causes divergence to "bleed" into that region as we've discussed. Similarly, some NVA can occur in the right-entrance region, which causes upper-level convergence to "bleed" into that region. The end result is the [two-quadrant model of a cyclonically curved jet streak](#), that you saw above.

[< Elevated Convection Coupled Jet Streams >](#)



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